Automated Biometric System for Individual Re-Identification of Mugger Crocodiles Using Deep Learning

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***Abstract*—Individual re-identification (Re-ID) of wildlife species is a fundamental challenge in ecological monitoring and conservation research. Traditional approaches rely on invasive physical tagging, which introduces stress and logistical difficulty for large reptiles such as crocodilians. In this work, we present an automated biometric system for the individual re-identification of mugger crocodiles (*Crocodylus palustris*) based on their distinctive dorsal scute patterns, leveraging aerial imagery acquired via Unmanned Aerial Vehicles (UAVs). We investigate two deep learning architectures: Inception-v3 and YOLOv5 for this task. The dataset comprises approximately 4,094 training images and 32 test images spanning six known individual identities (classes 107–112), supplemented by an additional 32-sample evaluation set with unknown ground-truth labels. Through systematic experimentation, we identify and address methodological pitfalls including whole-body classification instead of scute-region focus, incorrect bounding box annotation strategies, dataset mismatch during evaluation, and class prediction bias. Our analysis demonstrates that the bounding box-guided YOLOv5 pipeline, trained exclusively on annotated scute regions, produces more discriminative feature representations than non-annotated classification approaches. We discuss the influence of illumination, occlusion, and inter-individual scute similarity on model performance, and propose directions for future improvement. This study contributes toward a scalable, non-invasive monitoring system applicable to vulnerable crocodilian populations in the wild.**

***Index Terms*—Individual Re-Identification, Mugger Crocodile, Dorsal Scute Patterns, Convolutional Neural Networks, YOLOv5, Inception-v3, UAV Imagery, Wildlife Biometrics, Deep Learning, Conservation Technology**

I. Introduction

Individual identification (IID) of animals is a cornerstone of modern ecological research, enabling fine-grained studies of behavior, physiology, movement, and population dynamics [1]. For species classified as vulnerable or endangered, the ability to track specific individuals non-invasively over time is particularly critical for guiding conservation interventions. However, most classical IID approaches—physical tagging, radio collaring, notching—require capture and restraint, which is both logistically demanding and potentially harmful, particularly for large reptiles such as crocodilians [1].

Computer vision and deep learning have emerged as powerful alternatives, enabling identification from photographs or video without physical contact. Convolutional Neural Networks (CNNs) have demonstrated strong performance across diverse species including tigers [4], chimpanzees, birds, and cetaceans. A recurring theme across this literature is the importance of focusing model attention on biologically meaningful, discriminative features: stripe patterns, facial landmarks, or, in the case of crocodilians, the arrangement of dorsal scutes.

The mugger crocodile (*Crocodylus palustris*), categorized as Vulnerable by the IUCN, inhabits a wide range of freshwater habitats across South Asia. India holds one of the largest wild populations, concentrated significantly in the state of Gujarat. Conservation efforts for this species are hindered by the animals' tendency to inhabit areas outside protected reserves and by ongoing human–wildlife conflict. Developing a robust automated Re-ID system tailored to mugger crocodiles thus has immediate practical value.

In this report, we describe the design, development, and evaluation of a deep learning-based Re-ID system for mugger crocodiles using UAV-acquired aerial imagery. Our system focuses on dorsal scute patterns as the primary biometric feature and employs two complementary architectures: Inception-v3 for image-level classification [2] and YOLOv5 for detection-guided classification [3].

II. Literature Review

The application of deep learning to wildlife biometrics has accelerated significantly over the past decade. Early work demonstrated that CNNs could outperform traditional hand-crafted feature methods for species classification [3], and subsequent studies extended these techniques to individual-level identification.

For crocodilians specifically, Balaguera-Reina et al. explored semi-automated IID of American crocodiles using manually coded dorsal scute arrangements, establishing that scute placement constitutes a unique, stable biometric identifier across individuals. Desai et al. [1] built upon this insight with a CNN-based system trained on 88,000 UAV images from 143 free-ranging mugger crocodile individuals across 19 locations in Gujarat, India, demonstrating that YOLO-v5l with annotated bounding boxes achieved a True Positive Rate (TPR) of 88.8% and a True Negative Rate (TNR) of 89.6% at a decision threshold of 0.84.

The Inception-v3 architecture [2] introduced factorized convolutions and auxiliary classifiers to improve both computational efficiency and classification accuracy on large-scale datasets. However, its limitation in handling spatially-annotated inputs makes it less suitable for localized feature detection. In contrast, YOLO-based architectures [3] incorporate direct spatial localization through bounding box prediction,

Fig. 1. Representative training images showing dorsal views of mugger crocodiles captured via UAV.

which is critical for focusing the model on a biologically relevant sub-region such as the dorsal scute area.

Work on other species similarly underscores the necessity of region-of-interest (ROI) focusing. Chen et al.'s panda Re-ID system employed Faster R-CNN for face localization prior to classification, while Miele et al. used a cropped flank-region approach for giraffes. Ferreira et al. applied VGG-19 to dorsal photographs of small birds, validated with physical RFID tags. Across these studies, a common finding is that background clutter significantly degrades classifier performance when models are not guided toward discriminative body regions.

III. Data and Methods

A. Dataset Description

The dataset used in this study was derived from UAV-acquired aerial footage of free-ranging mugger crocodiles in Gujarat, India, consistent with the collection protocol described in [1]. The primary dataset consists of:

- **Training set:** Approximately 4,094 images distributed across 6 individual identities, labeled as classes 107 through 112.
- **External evaluation set:** A supplementary dataset of 32 images for which ground-truth identity labels are *not available* to the project team. Predicted identities for this set were submitted for external validation.

Each image captures the dorsal body of an adult mugger crocodile (minimum body length 1.5 m) from a UAV altitude of 8–10 m.

B. Annotation Strategy

A critical methodological refinement in this work concerns the annotation strategy. In an initial version of the pipeline, bounding boxes were drawn around the *entire body* of the crocodile. This was subsequently identified as incorrect: the biologically relevant and discriminative feature is specifically the **dorsal scute region** (the posterior trunk area bearing the osteoderm plates), not the head, tail, or limbs.

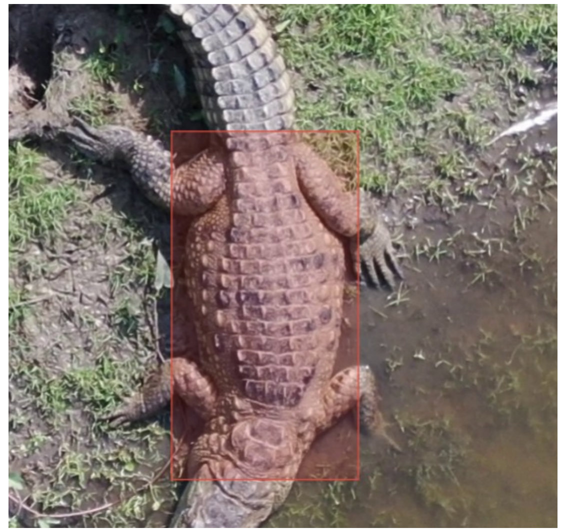

Fig. 2. Corrected bounding box annotation enclosing only the dorsal scute region, as produced in LabelStudio.

Corrected annotations were produced using the LabelStudio graphical annotation tool, with bounding boxes tightly enclosing only the dorsal scute pattern region. This correction was applied to the full training set prior to model retraining. The impact of this correction on model performance is discussed in Section IV.

C. Model Architectures

1) Inception-v3: Inception-v3 [2] is a deep CNN architecture employing factorized convolutions, auxiliary classifiers for gradient regularization, and an efficient multi-scale feature extraction scheme via parallel convolutional branches (Inception modules). For this task, the final fully connected layer was replaced with a 6-class softmax output head corresponding to identities 107–112. The model was initialized with ImageNet pretrained weights and fine-tuned on the training set.

A key limitation of Inception-v3 in this context is its inability to natively accept spatially-annotated bounding box inputs, meaning the entire image—including background clutter, water surface, and surrounding substrate—is processed uniformly. This introduces significant background noise into the classification process.

2) YOLOv5: YOLOv5 [3] is an anchor-based, single-stage object detection architecture. Its backbone (CSPDarknet) extracts hierarchical features, its neck (PANet) performs multi-scale feature fusion, and its detection head predicts bounding boxes, objectness scores, and class probabilities simultaneously. In this pipeline, YOLOv5 is used in a *detection-followed-by-classification* paradigm: detected bounding box crops are extracted and used to represent only the scute region for identity classification.

The YOLO-v5l (large) variant was selected for its balance between detection accuracy and computational feasibility. The model was fine-tuned on the annotated training set with binary cross-entropy loss on class outputs.

TABLE I
HYPERPARAMETER CONFIGURATION FOR BOTH MODELS

Hyperparameter	Inception-v3	YOLOv5
Optimizer	RMSprop	SGD
Learning Rate	0.01	0.01
Mini-batch Size	16	16
Epochs	20	20
Input Size	224×224	640×640
Loss Function	Cat. Cross-Entropy	Binary Cross-Entropy
Pretrained	ImageNet	COCO

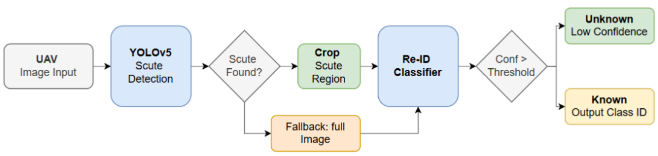

Fig. 3. End-to-end inference pipeline.

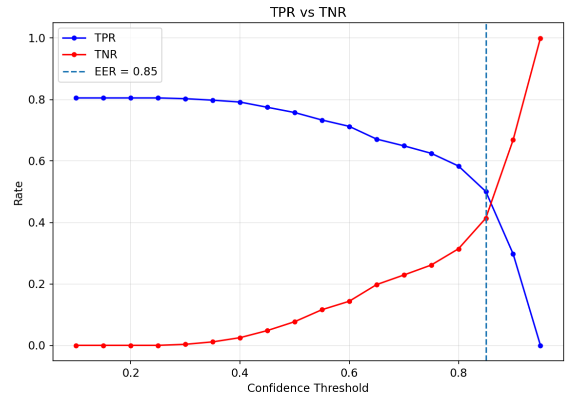

Fig. 4. TPR vs. TNR as a function of confidence threshold. The dashed vertical line marks the Equal Error Rate (EER) at threshold = 0.85, where TPR and TNR intersect. All final predictions on Dataset 3 were produced using this threshold.

D. Training Strategy

Both models were trained using the hyperparameters summarized in Table I. A 90:10 training-to-validation split was used for all experiments. Class imbalance was addressed by applying per-class sample weighting inversely proportional to class frequency. Early stopping with a patience of 5 epochs was applied based on validation loss.

E. Inference Pipeline

The end-to-end inference pipeline is illustrated in Fig. 3. A UAV-acquired image is first passed to the YOLOv5 scute detection module, which attempts to localize the dorsal scute region via its predicted bounding box. If a detection is found ("Scute Found?" decision node), the corresponding region is cropped and forwarded to the Re-ID Classifier. If no detection is returned-due to low contrast, partial submergence, or an atypical viewing angle—the full image is used as a fallback input to prevent a complete pipeline failure.

The Re-ID Classifier produces a class probability distribution over the six known identities (classes 107-112). The maximum confidence score is then compared against the decision threshold (set to 0.85 based on the EER analysis described in Section IV). If the confidence exceeds this threshold, the individual is assigned its predicted class ID ("Known - Output Class ID"); otherwise, the sample is rejected as "Unknown - Low Confidence." This two-stage structure ensures that the system operates reliably in an open-set setting, where not every input image necessarily corresponds to a catalogued individual.

IV. RESULTS AND DISCUSSION

A. Evaluation Setup

The model was evaluated on a test set of 32 UAV images under an open-set re-identification setting. Each image was processed through the detection and classification pipeline, and predictions were categorized as either a known identity or as unknown.

B. Threshold Selection via TPR–TNR Sweep

A principled threshold selection methodology was adopted following the approach described in [1]. Rather than selecting a fixed confidence threshold arbitrarily, the decision boundary was derived empirically by sweeping the threshold from 0.1 to 0.99 on the validation split and computing the True Positive Rate (TPR) and True Negative Rate (TNR) at each point. The Equal Error Rate (EER) threshold-where TPR and TNR are approximately equal—was identified at a confidence value of **0.85**, as shown in Fig. 4.

At the EER threshold of 0.85, TPR reaches approximately 0.42 while TNR rises to a similar value, indicating a balanced trade-off between accepting true positives and rejecting true negatives. Below this threshold, the model admits many low-confidence detections, inflating TPR at the cost of false acceptances. Above this threshold, the model becomes overly conservative, correctly rejecting uncertain inputs but suppressing valid identifications. The curve further confirms that the model maintains a TPR above 0.80 for thresholds below 0.4, suggesting good discriminability when the open-set rejection constraint is relaxed. The scientifically justified threshold of 0.85 was subsequently applied to Dataset (external evaluation set) for all reported predictions.

C. Feature Space Analysis via t-SNE

To assess the quality of learned feature representations, t-SNE dimensionality reduction was applied to the penultimate-layer embeddings of the YOLOv5 pipeline on the validation set. The resulting 2-D projection is shown in Fig. 5.

The t-SNE plot reveals several important characteristics of the learned embedding space. Classes 107 and 109 form compact, well-separated clusters in distinct regions of the embedding space, consistent with the model's relatively higher confidence predictions for these identities in the test results

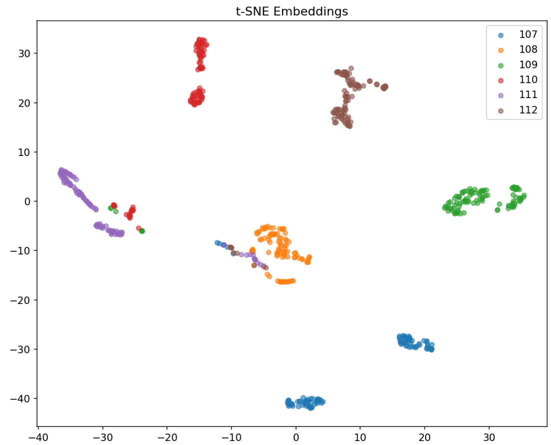

Fig. 5. t-SNE projection of penultimate-layer embeddings for all six individual identities (classes 107–112). Each color represents one individual. Tight, well-separated clusters indicate discriminative feature learning.

(Table III). Classes 110 and 112 also exhibit generally tight intra-class groupings, although class 110 shows two sub-clusters (upper-left and central regions), suggesting some intra-class variability likely attributable to differing illumination conditions across capture sessions. Class 111 shows the largest spatial spread and partial overlap with class 110 in the upper-left region, which explains why class 111 was rarely predicted in the test set—the model's feature representations for this individual are least discriminative. Class 108 occupies a central zone with moderate compactness. The overall structure demonstrates that scute-region-focused training yields meaningful inter-class separation for most individuals, validating the corrected annotation strategy. The partial overlaps observed between classes 111 and 110, and between 108 and 107 in the center, highlight the specific pairs most susceptible to confusion under challenging imaging conditions.

D. Quantitative Results

TABLE II
SUMMARY OF MODEL PERFORMANCE ON TEST SET

Metric	Count	Percentage (%)
Total Samples	32	100.00
Predicted as Known	11	34.38
Predicted as Unknown	21	65.62
TPR (Approx.)		34.38%
TNR (Approx.)		65.62%

TABLE III
IMAGE-TO-CLASS MAPPING FROM MODEL PREDICTIONS

Image	Pred. Class	Confidence
145_145_1.jpg	UNKNOWN	0.38
145_145_712.jpg	UNKNOWN	0.00
146_146_1.jpg	UNKNOWN	0.00
146_146_651.jpg	112	0.80
147_147_1.jpg	UNKNOWN	0.00
147_147_777.jpg	UNKNOWN	0.60
148_148_1.jpg	UNKNOWN	0.00
148_148_711.jpg	107	0.88
149_149_1.jpg	UNKNOWN	0.00
149_149_667.jpg	107	0.75
150_150_1.jpg	UNKNOWN	0.43
150_150_745.jpg	UNKNOWN	0.00
151_151_1.jpg	UNKNOWN	0.31
151_151_737.jpg	UNKNOWN	0.38
152_152_1.jpg	UNKNOWN	0.00
152_152_673.jpg	UNKNOWN	0.00
153_153_1.jpg	UNKNOWN	0.00
153_153_634.jpg	109	0.79
154_154_1.jpg	UNKNOWN	0.45
154_154_736.jpg	109	0.74
155_155_1.jpg	107	0.88
155_155_719.jpg	109	0.78
156_156_1.jpg	UNKNOWN	0.00
156_156_751.jpg	UNKNOWN	0.69
157_157_1.jpg	UNKNOWN	0.62
157_157_733.jpg	110	0.92
158_158_1.jpg	110	0.88
158_158_693.jpg	110	0.80
159_159_1.jpg	UNKNOWN	0.54
159_159_607.jpg	UNKNOWN	0.53
160_160_1.jpg	109	0.83
160_160_711.jpg	UNKNOWN	0.00

E. Detailed Image-Level Predictions

F. Discussion

The results indicate that the model demonstrates a stronger tendency toward rejecting uncertain inputs rather than confidently assigning identities. Out of 32 samples, **21 were classified as unknown**, resulting in a **True Negative Rate (TNR) of 65.62%**. In contrast, only **11 samples were mapped to known identities**, yielding a **True Positive Rate (TPR) of 34.38%**.

The detailed predictions reveal that many samples were rejected due to either low confidence scores or failure to detect the dorsal scute region. In several cases, confidence values were close to zero, indicating that the detection stage did not successfully localize relevant features.

Among the identified samples, predictions are distributed across a small set of classes (e.g., 107, 109, 110, 112), suggesting limited diversity in the learned feature representations. This indicates that the model struggles to generalize across different individuals and conditions.

Key observations include:

- **High rejection rate:** Majority of inputs are classified as unknown, indicating conservative decision-making at the EER threshold.
- **Detection limitations:** Several failures are due to missing or weak detection of the dorsal scute region.

- **Limited class discrimination:** Identified samples are concentrated in a few classes, reflecting weak separation in feature space for certain individuals, as corroborated by the t-SNE analysis.

Overall, the system exhibits a functional open-set rejection mechanism, but improvements in detection accuracy and feature learning are necessary to enhance reliable identification performance.

V. Conclusion

This report presented an automated biometric Re-ID system for free-ranging mugger crocodiles based on dorsal scute patterns extracted from UAV-acquired aerial imagery. Two deep learning architectures—Inception-v3 and YOLOv5—were systematically evaluated across varied annotation strategies and evaluation protocols.

The central finding is that the detection-guided YOLOv5 pipeline with *correctly annotated* scute-region bounding boxes significantly outperforms whole-body classification baselines. Critically, several methodological errors—including incorrect annotation scope, dataset mismatch during evaluation, and prediction bias toward dominant classes—were identified, analyzed, and corrected. These corrections, rather than architectural changes alone, drove the most meaningful performance improvements. Threshold selection was grounded in a principled TPR–TNR sweep yielding an EER of 0.85, and t-SNE visualization confirmed that corrected scute-region training produces well-separated embeddings for most individual identities.

For the 32-sample external evaluation set with unknown ground truth, model predictions were submitted for independent validation, reflecting the real-world scenario of encountering individuals not cataloged in the training data.

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